**Supplementary Table S1** OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE ® Daily and Ovid MEDLINE ® 1946 to August 2019 Search Strategy

1. Venous thromboembolism/
2. exp Venous Thrombosis/
3. exp Pulmonary Embolism/
4. dvt.mp.
5. pulmonary embolism.ti,ab.
6. ((ven* or vein*) adj2 thromb*).ti,ab.
**7. or/1–6**
8. exp Angiogenesis Inhibitors/
9. exp Anticarcinogenic Agents/
10. exp Antimetabolites, antineoplastic/
11. exp Antimitotic Agents/
12. exp Antineoplastic agents, alkylating/
13. exp Antineoplastic agents, immunological/
14. exp Topoisomerase inhibitors/
15. immunotherap*.ti,ab
16. systemic therapy.ti,ab.
17. chemo*.ti,ab.
18. antineoplastic.ti,ab.
19. targeted therap*.ti,ab.
20. targetted therap*.ti,ab.
21. biologic* agent*.ti,ab.
22. antitumor.ti,ab.
23. antitumor.ti,ab.
**24. or/8–23**
25. exp Neoplasms/
26. cancer*.ti,ab.
27. malig*.ti,ab.
28. neoplas*.ti,ab.
29. oncol*.ti,ab.
30. tumor*.ti,ab.
31. tumor*.ti,ab
**32. or/25–31**
33. exp Risk Assessment/
34. Incidence/
35. Incidence Studies/
36. Risk Factors/
37. associat*.ti,ab.
38. validat*.ti,ab.
39. incidence*.ti,ab.
40. predict*.ti,ab.
41. risk factor*.ti,ab.
42. prognos*.ti,ab.
**43. or/33–42**
**44. 7 and 24 and 32 and 43**
45. limit 38 to ("review" or "scientific integrity review" or "systematic review" or systematic reviews as topic)
46. exp Animals/ not exp Humans/
47. 44 not 45
**48. 47 not 46**

**Supplementary Table S2** Embase 1974 to August 2019 Search Strategy

1. exp Venous thromboembolism/
2. exp Vein Thrombosis/
3. exp Lung Embolism/
4. dvt.ti,ab.
5. pulmonary embolism.ti,ab.
6. ((ven* or vein*) adj2 thromb*).ti,ab.
**7. or/1–6**
8. Antineoplastic Agent/
9. immunotherap*.ti,ab
10. systemic therapy.ti,ab.
11. chemo*.ti,ab.
12. antineoplastic.ti,ab.
13. targeted therap*.ti,ab.
14. targetted therap*.ti,ab.
15. biologic* agent*.ti,ab.
16. antitumor.ti,ab.
17. antitumor.ti,ab.
**18. or/8–17**
19. exp Malignant Neoplasm/
20. cancer*.ti,ab.
21. malig*.ti,ab.
22. neoplas*.ti,ab.
23. oncol*.ti,ab.
24. tumor*.ti,ab.
25. tumor*.ti,ab
**26. or/19–25**
27. exp Risk Assessment/
28. Incidence/
29. Risk Factor/
30. validat*.ti,ab.
31. incidence*.ti,ab.
32. predict*.ti,ab.
33. risk factor*.ti,ab.
34. prognos*.ti,ab.
**35. or/27–34**
**36. 7 and 18 and 26 and 35**
37. limit 36 to ("review")
38. exp Animal/ not exp Human/
39. 36 not 37
**40. 39 not 38**

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**Supplementary Table S3** Cochrane Controlled Register of Trials (CENTRAL) Database Conception to August 2019 Search Strategy

| |
|---|
| 1. Venous thromboembolism/ |
| 2. exp Venous Thrombosis/ |
| 3. exp Pulmonary Embolism/ |
| 4. dvt.mp. |
| 5. pulmonary embolism.ti,ab. |
| 6. ((ven* or vein*) adj2 thromb*).ti,ab. |
| **7. or/1–6** |
| 8. exp Angiogenesis Inhibitors/ |
| 9. exp Anticarcinogenic Agents/ |
| 10. exp Antimetabolites, antineoplastic/ |
| 11. exp Antimitotic Agents/ |
| 12. exp Antineoplastic agents, alkylating/ |
| 13. exp Antineoplastic agents, immunological/ |
| 14. exp Topoisomerase inhibitors/ |
| 15. ssociateapy*.ti,ab |
| 16. systemic therapy.ti,ab. |
| 17. chemo*.ti,ab. |
| 18. antineoplastic.ti,ab. |
| 19. targeted therap*.ti,ab. |
| 20. ssociat therap*.ti,ab. |
| 21. biologic* agent*.ti,ab. |
| 22. antitumor.ti,ab. |
| 23. antitumor.ti,ab. |
| **24. or/8–23** |
| 25. exp Neoplasms/ |
| 26. cancer*.ti,ab. |
| 27. malig*.ti,ab. |
| 28. neoplas*.ti,ab. |
| 29. oncol*.ti,ab. |
| 30. tumor*.ti,ab. |
| 31. tumor*.ti,ab |
| **32. or/25–31** |
| 33. exp Risk Assessment/ |
| 34. Incidence/ |
| 35. Risk Factors/ |
| 36. associate*.ti,ab. |
| 37. validat*.ti,ab. |
| 38. incidence*.ti,ab. |
| 39. predict*.ti,ab. |
| 40. risk factor*.ti,ab. |
| 41. prognos*.ti,ab. |
| **42. or/33–41** |
| **43. 7 and 24 and 32 and 42** |

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**Supplementary Table S4** Risk of bias assessment using the QUIPS Tool

| Author and year | Study participation | Study attrition | Prognostic factor measurement | Outcome measurement | Study confounding | Statistical analysis and reporting | Overall study quality |
|---|---|---|---|---|---|---|---|
| Abdel-Razeq 2018 | high risk | low risk | moderate risk | low risk | low risk | low risk | high risk |
| Arpaia 2009 | low risk | low risk | low risk | low risk | high risk | moderate risk | high risk |
| Di Nisio 2019 | high risk | high risk | moderate risk | low risk | high risk | low risk | high risk |
| Ferroni (GFR) 2014* | low risk | moderate risk | moderate risk | low risk | moderate risk | high risk | high risk |
| Ferroni (MPV) 2014* | high risk | high risk | moderate risk | low risk | moderate risk | high risk | high risk |
| Ferroni 2015* | high risk | low risk | moderate risk | low risk | low risk | moderate risk | high risk |
| Ferroni 2016 | low risk | low risk | moderate risk | low risk | high risk | moderate risk | high risk |
| Gerotziafas 2017 | low risk | moderate risk | low risk | low risk | moderate risk | low risk | moderate risk |
| Khorana 2005** | high risk | moderate risk | low risk | low risk | low risk | high risk | high risk |
| Khorana 2008** | high risk | moderate risk | moderate risk | low risk | low risk | moderate risk | high risk |
| Roselli 2013* | low risk | high risk | moderate risk | low risk | high risk | high risk | high risk |
| Tafur 2015 | high risk | high risk | low risk | low risk | moderate risk | high risk | high risk |
| van Es 2017*** | high risk | high risk | low risk | low risk | moderate risk | high risk | high risk |
| van Es 2018*** | high risk | high risk | low risk | low risk | high risk | high risk | high risk |
| Vergati 2013* | low risk | low risk | low risk | low risk | moderate risk | high risk | high risk |

*Study conducted by same research group
**Study conducted by same research group
***Study conducted by same research group
The domain with the highest risk of bias was used to determine the overall study quality.

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**Supplementary Table S5** Result of multivariable analysis of prognostic factors for VTE

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| **Patient Factors** | | | | | | |
| Age | Vergati 2013 | 486 | >70 vs ≤70 years | HR | **2.43 (1.16–5.09)** | Pos |
| | Arpaia 2009 | 124 | ND | HR | 1.04 (0.9–1.09) | Pos |
| | van Es 2018*** | 684 | per 1 year increase | SHR | 1.00 (0.96–1.03) | N/A |
| | Roselli 2013* | 505 | >65 vs ≤65 years | HR | 0.99 (0.49–2.02) | Neg |
| | Ferroni 2014 (MPV)* | 589 | ND | HR | 1.00 (0.97–1.02) | N/A |
| | Ferroni 2014 (GFR)* | 322 | >65 vs ≤65 years | OR | 0.97 (0.38–2.46) | Pos |
| | Ferroni 2015* | 380 | >65 vs ≤65 years | HR | 0.78 (0.35–1.74) | Neg |
| | Ferroni 2016* | 297 | >65 vs ≤65 years | HR | 1.97 (0.82–4.71) | Pos |
| Gender | Abdel-Razeq 2018 | 1677 | vs male | OR | **1.732 (1.152–2.605)** | Pos |
| | Vergati 2013 | 486 | vs male | HR | 0.60 (0.28–1.29) | Neg |
| | van Es 2018*** | 648 | vs male | SHR | 1.05 (2.13–0.53) | Pos |
| | Roselli 2013* | 505 | ND | HR | 0.79 (0.39–1.59) | Neg |
| | Ferroni 2014 (MPV)* | 589 | ND | HR | 1.15 (0.57–2.33) | Pos |
| | Ferroni 2014 (GFR)* | 322 | ND | OR | 1.26 (0.48–3.36) | Pos |
| | Ferroni 2015* | 380 | vs male | HR | 1.38 (0.57–3.35) | Pos |
| | Ferroni 2016* | 297 | vs male | HR | 1.40 (0.57–3.43) | Pos |
| BMI | Di Nisio 2019 | 776 | >35 vs ≤35 | SHR | 1.6 (0.50–5.4)[a] | Pos |
| | | | | SHR | 1.8 (0.55–5.9)[b] | Pos |
| | Khorana 2008** | 2701 | >35 vs ≤35 | OR | **2.5 (1.3–4.7)** | Pos |
| | Ferroni 2015* | 380 | >35 vs ≤35 | HR | 0.46 (0.03–7.18) | Neg |
| | Roselli 2013* | 505 | ND | HR | 1.04 (0.97–1.11) | Pos |
| ECOG Status | Di Nisio 2019 | 776 | ≥2 vs <2 points | SHR | 1.5 (0.46–4.6)[c] | Pos |
| | van Es 2017*** | 876 | ≥2 vs <2 points | SHR | 1.2 (0.5–2.9)[c] | Pos |
| | Vergati 2013 | 486 | 2 vs <2 points | HR | **6.63 (3.13–14.0)** | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | **2.25 (1.18–4.27)** | Pos |
| | Roselli 2013* | 505 | | HR | **4.26 (2.15–8.43)** | Pos |
| | Ferroni 2014 (GFR)* | 322 | | OR | 1.72 (0.65–4.52) | Pos |
| | Ferroni 2015* | 380 | | HR | **2.57 (1.34–4.93)** | Pos |

(*Continued*)



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**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| | Ferroni 2016* | 297 | | HR | **4.13 (1.84–9.27)** | Pos |
| Khorana Score | Abdel-Razeq 2018 | 1677 | ≥3 vs <3 points | OR | 1.387 (0.842–2.285) | Pos |
| | Di Nisio 2019 | 776 | ≥2 vs <2 points | SHR | 1.8 (0.95–3.4)[d] | Pos |
| | Vergati 2013 | 486 | ND | HR | 1.34 (0.7–2.58) | Pos |
| | van Es 2018*** | 648 | per 1 point increase | SHR | 1.008 (0.768–1.32) | Pos |
| | Ferroni 2016* | 297 | ND | HR | **4.13 (1.84–9.27)** | Pos |
| | Ferroni 2014 (MPV)* | 589 | ND | HR | 1.15 (0.62–2.15) | Pos |
| | Ferroni 2014 (GFR)* | 322 | ND | OR | 2.42 (0.91–6.48) | Pos |
| Previous VTE | Di Nisio 2019 | 776 | Any VTE that occurred 6+ months before enrollment | SHR | **17 (5.2–52)**[d] | Pos |
| Vascular/lymphatic macroscopic compression | Di Nisio 2019 | 776 | Vascular/lymphatic macroscopic compression by tumor, detected by venous compression ultrasound (B-mode imaging) | SHR | 1.7 (0.60–5.0)[d] | Pos |
| Cardiovascular comorbidities | Gerotziafas 2017 | 1023 | At least 2 of: personal history of peripheral artery disease, ischemic stroke, coronary artery disease, hypertension, hyperlipidemia, diabetes, obesity | OR | **5.18 (1.10–13.40)** | Pos |
| **Tumour Factors** | | | | | | |
| Tumor site (High risk vs low risk) | Di Nisio 2019 | 776 | High Risk (Lung, Lymphoma, Gynecologic, Genitourinary Except Prostate) vs Low Risk (breast, colorectal, head-neck) | SHR | 1.9 (0.71–4.9)[a] | Pos |
| | | | | SHR | 2.2 (0.83–6.0)[b] | Pos |
| | | | | SHR | 1.8 (0.68–4.7)[c] | Pos |
| | Khorana 2008** | 2701 | | OR | 1.5 (0.9–2.7) | Pos |
| | van Es 2017*** | 876 | | SHR | 1.10 (0.59–2.10)[a] | Pos |
| | | | | SHR | 1.00 (0.53–1.80)[e] | N/A |
| | | | | SHR | 0.66 (0.33–1.3)[b] | Neg |
| | | | | SHR | 1.1 (0.58–2)[c] | Pos |

**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| Tumor site (Very high risk vs low risk) | Di Nisio 2019 | 776 | Very High Risk (Stomach, Pancreas) vs Low Risk (breast, colorectal, head-neck) | SHR | 1.8 (0.85–3.9)[a] | Pos |
| | | | | SHR | 1.7 (0.72–3.8)[b] | Pos |
| | | | | SHR | 1.8 (0.85–3.9)[c] | Pos |
| | Khorana 2008** | 2701 | | OR | **4.3 (1.2–15.6)** | Pos |
| | Ferroni 2015* | 380 | | HR | 1.32 (0.63–2.80) | Pos |
| | Khorana 2005** | 3003 | | OR | **3.88 (1.43–10.05)** | Pos |
| | van Es 2017*** | 876 | | SHR | 1.40 (0.72–2.90)[a] | Pos |
| | | | | SHR | 1.50 (0.72–2.90)[e] | Pos |
| | | | | SHR | 1.10 (0.54–2.20)[b] | Pos |
| | | | | SHR | 1.50 (0.72–2.90)[c] | Pos |
| Tumor site (Other) | Abdel-Razeq 2018 | 1677 | Gastric vs others | OR | **3.377 (1.759–6.483)** | Pos |
| | Vergati 2013 | 486 | ND | HR | 0.97 (0.84–1.10) | Neg |
| | Ferroni 2014 (MPV)* | 589 | ND | HR | 1.07 (0.76–1.51) | Pos |
| | Ferroni 2014 (GFR)* | 322 | ND | OR | 1.63 (0.94–2.82) | Pos |
| Tumor stage | Di Nisio 2019 | 776 | Metastatic vs Non-metastatic | SHR | 1.1 (0.52–2.30)[d] | Pos |
| | van Es 2018*** | 648 | | SHR | 1.26 (0.61–2.61) | Pos |
| | Vergati 2013 | 486 | Non-metastatic vs Metastatic | HR | 0.88 (0.31–2.48) | Neg |
| | Roselli 2013* | 505 | Metastatic vs Non-metastatic | HR | 1.87 (0.76–4.65) | Pos |
| | Arpaia 2009 | 124 | | HR | 1.25 (0.35–4.37) | Pos |
| | Abdel-Razeq 2018 | 1677 | | OR | **1.67 (1.05–2.63)** | Pos |
| | Ferroni 2015* | 380 | | HR | 1.01 (0.45–2.25) | Pos |
| | Ferroni 2014 (GFR)* | 322 | | OR | 1.32 (0.49–3.55) | Pos |
| | Ferroni 2014 (MPV)* | 589 | ND | HR | 1.15 (0.93–1.41) | Pos |
| | Ferroni 2016* | 297 | Relapsing/metastatic | HR | 1.41 (0.53–3.75) | Pos |
| | Gerotziafas 2017 | 1023 | Advanced stage or metastatic | OR | 1.93 (0.92–2.64) | Pos |
| Time since cancer diagnosis | Gerotziafas 2017 | 1023 | ≤6 months since diagnosis | OR | **4.10 (2.10–7.98)** | Pos |

(*Continued*)

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**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| **Biomarkers** | | | | | | |
| Prechemotherapy hemoglobin | Di Nisio 2019 | 776 | Prechemotherapy hemoglobin <100 g/L or use of erythropoietin stimulating agents | SHR | 1.5 (0.58–4.1)[a] | Pos |
| | | | | SHR | 1.7 (0.63–4.6)[b] | Pos |
| | | | | SHR | 1.6 (0.59–4.1)[c] | Pos |
| | Khorana 2005** | 3003 | | OR | **1.83 (1.07–3.14)** | Pos |
| | van Es 2017*** | 876 | | SHR | 0.76 (0.32–1.80)[a] | Neg |
| | | | | SHR | 0.64 (0.27–1.60)[e] | Neg |
| | | | | SHR | 0.70 (0.28–1.70)[b] | Neg |
| | | | | SHR | 0.76 (0.31–1.80)[c] | Neg |
| | Khorana 2008** | 876 | | OR | **2.4 (1.4–4.2)** | Pos |
| | Ferroni 2015* | 380 | Haemoglobin <100 g L^-11 | HR | 1.54 (0.43–5.50) | Pos |
| Prechemotherapy white blood cell count | Di Nisio 2019 | 776 | Prechemotherapy white blood cell count >11 × 10^9/L | SHR | 1.7 (0.75–3.6)[a] | Pos |
| | | | | SHR | 1.81 (0.80–4.1)[b] | Pos |
| | | | | SHR | 1.7 (0.76–3.6)[c] | Pos |
| | Khorana 2008** | 2701 | | OR | **2.2 (1.2–4)** | Pos |
| | Ferroni 2015* | 380 | | HR | 0.79 (0.23–2.65) | Neg |
| | van Es 2017*** | 876 | | SHR | 0.87 (0.42–1.80)[a] | Neg |
| | | | | SHR | 0.82 (0.39–1.80)[e] | Neg |
| | | | | SHR | 0.77 (0.36–1.7)[b] | Neg |
| | | | | SHR | 0.86 (0.41–1.8)[c] | Neg |
| Prechemotherapy platelet count | Di Nisio 2019 | 776 | ≥ 350 × 10^9/L | SHR | 0.97 (0.45–2.1)[a] | Neg |
| | | | | SHR | 0.82 (0.38–1.8)[b] | Neg |
| | | | | SHR | 0.98 (0.46–2.1)[c] | Neg |
| | Khorana 2008** | 2701 | | OR | **1.8 (1.1–3.2)** | Pos |
| | Gerotziafas 2017 | 1023 | | OR | **2.53 (1.35–4.74)** | Pos |
| | van Es 2017*** | 876 | | SHR | 1.40 (0.80–2.40)[a] | Pos |
| | | | | SHR | 1.30 (0.71–2.20)[e] | Pos |
| | | | | SHR | 1.30 (0.74–2.30)[b] | Pos |
| | | | | SHR | 1.40 (0.80–2.40)[c] | Pos |
| | Khorana 2005** | 3003 | | OR | **2.81 (1.63–4.93)** | Pos |
| | Ferroni 2015* | 380 | >350 × 10^9/L | HR | 0.18 (0.03–1.02) | Neg |
| | Ferroni 2014 (MPV)* | 589 | ≤7.3 fL or >7.3 fL | HR | **2.32 (1.03–5.23)** | Pos |

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**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| Low protein C activity (%) | Tafur 2015 | 241 | Dichotomized as ≤ 118% (lower quartile) and >118% | HR | **2.5 (1.1–5.5)** | Pos |
| High factor VIII activity (%) | Tafur 2015 | 241 | Dichotomized as ≤ 261% and >261% | HR | **3.0 (1.1–8.0)** | Pos |
| ThromboPath change (%) | Roselli 2013* | 505 | Decreasing (-6% cutoff) or not | HR | **2.39 (1.20–4.78)** | Pos |
| D-dimer levels | van Es 2018*** | 648 | Per 1 ug/L [log transformed] increase | SHR | **1.91 (1.40–2.62)** | Pos |
| | Arpaia 2009 | 124 | >650 ng/mL | HR | **4.04 (1.22–13.3)** | Pos |
| | van Es 2017*** | 876 | >1.44 g/L | SHR | **2.40 (1.30–4.40)**[e] | Pos |
| | Vergati 2013 | 486 | Negative vs positive (cutoff of 280 ng/mL) | HR | 3.11 (0.83–11.7) | Pos |
| Fibrin generation test | van Es 2018*** | 648 | High versus low activity | SHR | 1.94 (0.99–3.80) | Pos |
| eGFR | Ferroni 2014 (GFR)* | 322 | eGFR categorized as normal or impaired based on the cutoff of normal being >90 mL/min per 1.73 m | OR | **3.44 (1.27–9.34)** | Pos |
| Platelet/lymphocyte ratio | Ferroni 2015* | 380 | Platelet/lymphocyte ratio >260 | HR | **2.72 (1.10–6.73)** | Pos |
| Neutrophil/lymphocyte ratio | Ferroni 2015* | 380 | Neutrophil/lymphocyte ratio >3 | HR | 1.37 (0.52–3.57) | Pos |
| **Treatment Factors** | | | | | | |
| Hospitalization | Gerotziafas 2017 | 1023 | Hospitalization for any non-surgical reason occurring within the last 3 months before assessment | OR | **5.41 (2.90–10.08)** | Pos |
| Adjuvant chemotherapy | Arpaia 2009 | 124 | vs first-line chemotherapy | HR | 0.69 (0.07–6.93) | Neg |
| Central venous catheter | Abdel-Razeq 2018 | 1677 | vs no catheter | OR | 1.098 (0.603–1.999) | Pos |
| | Gerotziafas 2017 | 1023 | vs no catheter | OR | **3.24 (1.56–6.72)** | Pos |
| *Cytotoxic Chemotherapy* | | | | | | |
| Gemcitabine | Di Nisio 2019 | 776 | vs other treatment | SHR | 0.70 (0.34–1.5)[b] | Neg |
| | van Es 2017*** | 876 | | SHR | **3.70 (1.80–7.60)**[b] | Pos |
| | Vergati 2013 | 486 | | HR | 0.42 (0.11–1.56) | Neg |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.66 (0.23–1.90) | Neg |
| | Ferroni 2015* | 380 | | HR | 0.45 (0.16–1.32) | Neg |
| | Ferroni 2016* | 297 | | HR | 1.78 (0.23–13.9) | Pos |

(*Continued*)

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**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| | Ferroni 2014 (GFR)* | 322 | | OR | 0.51 (0.14–1.82) | Neg |
| Platinum | Vergati 2013 | 486 | vs other treatment | HR | **2.51 (1.08–5.83)** | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 1.25 (0.53–2.95) | Pos |
| | Di Nisio 2019 | 776 | | SHR | **2.5 (1.1–6.1)**[b] | Pos |
| | van Es 2017*** | 876 | | SHR | **2.8 (1.4–5.6)**[b] | Pos |
| | Roselli 2013* | 505 | | HR | **2.19 (1.03–4.68)** | Pos |
| | Ferroni 2016* | 297 | | HR | **4.86 (1.27–18.60)** | Pos |
| | Ferroni 2015* | 380 | | HR | 0.86 (0.35–2.13) | Neg |
| Fluoropyrimidine | Ferroni 2016* | 297 | vs other treatment | HR | 3.71 (0.61–22.36) | Pos |
| | Roselli 2013* | 505 | | HR | 1.57 (0.69–3.55) | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.89 (0.35–2.25) | Neg |
| | Ferroni 2014 (GFR)* | 322 | | OR | 2.27 (0.36–14.30) | Pos |
| | Ferroni 2015* | 380 | | HR | 0.54 (0.16–1.85) | Neg |
| Irinotecan | Ferroni 2016* | 297 | vs other treatment | HR | 1.44 (0.36–5.70) | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.85 (0.25–2.897) | Neg |
| | Ferroni 2015* | 380 | | HR | 0.85 (0.16–4.45) | Neg |
| Anthracycline | Ferroni 2016* | 297 | vs other treatment | HR | 0.18 (0.02–1.87) | Neg |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.40 (0.08–1.95) | Neg |
| | Ferroni 2014 (GFR)* | 322 | | OR | 0.33 (0.04–1.50) | Neg |
| | Gerotziafas 2017 | 1023 | | OR | **2.33 (1.02–5.33)** | Pos |
| | Ferroni 2015* | 380 | | HR | 1.33 (0.31–5.68) | Pos |
| Docetaxel | Ferroni 2016* | 297 | vs other treatment | HR | **7.47 (1.18–47.4)** | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.92 (0.30–2.88) | Neg |
| | Ferroni 2014 (GFR)* | 322 | | OR | 0.25 (0.04–1.50) | Neg |
| | Ferroni 2015* | 380 | | HR | 0.71 (0.21–2.42) | Neg |
| Pemetrexed | Ferroni 2014 (MPV)* | 589 | vs other treatment | HR | 0.75 (0.19–2.94) | Neg |
| | Ferroni 2014 (GFR)* | 322 | | OR | 0.11 (0.01–1.14) | Neg |
| | Ferroni 2015* | 380 | | HR | 0.43 (0.11–1.76) | Neg |

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**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| *Biological Therapy* | | | | | | |
| Bevacizumab | Ferroni 2016* | 297 | vs other treatment | HR | **3.56 (1.12–11.40)** | Pos |
| | Roselli 2013* | 505 | | HR | 1.21 (0.43–3.40) | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 2.59 (0.84–1.15) | Pos |
| | Vergati 2013 | 486 | | HR | 2.49 (0.84–7.32) | Pos |
| | Ferroni 2014 (GFR)* | 322 | | OR | 1.00 (0.10–9.87) | N/A |
| | Ferroni 2015* | 380 | | HR | **4.54 (1.27–16.2)** | Pos |
| Herceptin | Ferroni 2014 (MPV)* | 589 | vs other treatment | HR | 0.96 (0.11–8.11) | Neg |
| | Ferroni 2015* | 380 | | HR | 0 (No events) | N/A |
| | Ferroni 2014 (GFR)* | 322 | | OR | 0 (No events) | N/A |
| Anti-tyrosine kinase inhibitors | Ferroni 2014 (MPV)* | 589 | vs other treatment | HR | 1.23 (0.13–11.5) | Pos |
| | Vergati 2013 | 486 | | HR | 1.75 (0.21–14.7) | Pos |
| | Ferroni 2015* | 380 | | HR | 6.29 (0.59–67.2) | Pos |
| *Other* | | | | | | |
| Endocrine/anti-hormonal therapy | Ferroni 2014 (MPV)* | 589 | vs other treatment | HR | 0.78 (0.09–6.80) | Neg |
| | Gerotziafas 2017 | 1023 | vs other treatment in women with breast cancer | OR | **6.40 (3.16–12.96)** | Pos |
| *Supportive drug use during chemotherapy* | | | | | | |
| Erythropoietin stimulating agents | Di Nisio 2019 | 776 | vs none | SHR | 1.5 (0.58–4.1)[a] | Pos |
| | | | | SHR | 1.7 (0.63–4.6)[b] | Pos |
| | | | | SHR | 1.6 (0.59–4.1)[c] | Pos |
| | Ferroni 2015* | 380 | | HR | 1.49 (0.34–6.66) | Pos |
| | Roselli 2013* | 505 | | HR | 1.33 (0.32–5.55) | Pos |
| | van Es 2017*** | 876 | | SHR | 0.76 (0.32–1.80)[a] | Neg |
| | | | | SHR | 0.64 (0.27–1.60)[e] | Neg |
| | | | | SHR | 0.70 (0.28–1.70)[b] | Neg |
| | | | | SHR | 0.76 (0.31–1.80)[c] | Neg |
| Prophylactic myeloid growth factors | Roselli 2013* | 505 | vs none | HR | 1.25 (0.31–5.08) | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.97 (0.76–1.23) | Neg |
| | Vergati 2013 | 486 | | HR | 0.60 (0.08–4.61) | Neg |
| | Ferroni 2014 (GFR)* | 322 | | OR | 1.18 (0.19–7.37) | Pos |
| | Ferroni 2015* | 380 | | HR | 1.00 (0.22–4.73) | Pos |

(*Continued*)

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**Supplementary Table S5** (*Continued*)

| Prognostic Factor | Studies | Sample size | Definition | Measure of association | Risk estimate[1] (95% CI) | Pos/neg association[2] |
|---|---|---|---|---|---|---|
| Corticosteroids | Roselli 2013* | 505 | vs none | HR | 1.18 (0.52–2.65) | Pos |
| | Ferroni 2014 (MPV)* | 589 | | HR | 0.98 (0.84–1.15) | Neg |
| | Vergati 2013 | 486 | | HR | 1.84 (0.86–3.94) | Pos |
| | Ferroni 2014 (GFR)* | 322 | | OR | 1.50 (0.48–4.66) | Pos |
| | Ferroni 2015* | 380 | | HR | 1.16 (0.46–2.88) | Pos |
| **Genetic factors** | | | | | | |
| VEGFA-1190G/A A/A polymorphism | Ferroni 2016* | 297 | vs none | HR | 1.24 (0.54–2.88) | Pos |
| VEGFA-1154G/A A/A polymorphism | Ferroni 2016* | 297 | vs none | HR | **0.21 (0.07–0.58)** | Neg |
| VEGFA-634G/C C/C polymorphism | Ferroni 2016* | 297 | vs none | HR | 0.82 (0.38–1.77) | Neg |

Abbreviations: ECOG, Eastern Cooperative Oncology Group; ND, Not Defined (in the original study).

*Study conducted by same research group.

**Study conducted by same research group.

***Study conducted by same research group.

[1]Statistically significant results are indicated in bold text.

[2]Positive association refers to a point estimate greater than 1 and a negative association refers to a point estimate less than 1.

[a]Adjusted by Khorana score variables (Pancreatic or gastric cancer; lung, ovarian, urogenital; prechemotherapy hemoglobin; prechemotherapy white blood cell count; prechemotherapy platelet count; Body mass index >35).

[b]Adjusted by PROTECHT score variables (Pancreatic or gastric cancer; lung, ovarian, urogenital; prechemotherapy hemoglobin; prechemotherapy white blood cell count; prechemotherapy platelet count; Body mass index >35; gemcitabine chemotherapy; platinum-based chemotherapy).

[c]Adjusted by CONKO score variables (Pancreatic or gastric cancer; lung, ovarian, urogenital; prechemotherapy hemoglobin; prechemotherapy white blood cell count; prechemotherapy platelet count; ECOG performance status ≥2).

[d]Adjusted by ONKOTEV score variables (Khorana score ≥2; metastasis; previous venous thromboembolism; vascular/lymphatic macroscopic compression.

[e]Adjusted by Vienna CATS score variables (Pancreatic or gastric cancer; lung, ovarian, urogenital; prechemotherapy hemoglobin; prechemotherapy white blood cell count; prechemotherapy platelet count; Body mass index >35; D-dimer; soluble P-selectin.

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